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Department of Computer Engineering

Salon Appointment & Management System

Presented By

Ayush Thakare (124B1B094)
Sarang Wasamwar (124B1B090)
Shubham Navale (124B1B118)

Guided By

Prof. Mrs. B. Mahalakshmi

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Introduction

SalonSnap is a comprehensive web-based salon management system designed to digitize and streamline salon operations. It provides a seamless platform for customers to book appointments online, make secure payments, and download digital receipts. For salon administrators, it offers a powerful dashboard to manage services, staff, appointments, payments, and customer inquiries. The system eliminates manual booking processes, reduces no-shows through automated reminders, and enhances customer experience with instant payment confirmation and receipt generation. Built using Node.js, MySQL, and Razorpay integration, SalonSnap bridges the gap between customers and salon businesses, making beauty services more accessible, organized, and efficient. The system also supports mobile app integration via Supabase, enabling cross-platform accessibility.

Problem Statment

Traditional salon management relies on manual booking systems, phone calls, and walk-ins, leading to issues like double bookings, long waiting times, lost customer records, and inefficient payment tracking. There is no centralized system for managing appointments, staff, services, and payments, resulting in poor customer experience and administrative chaos.

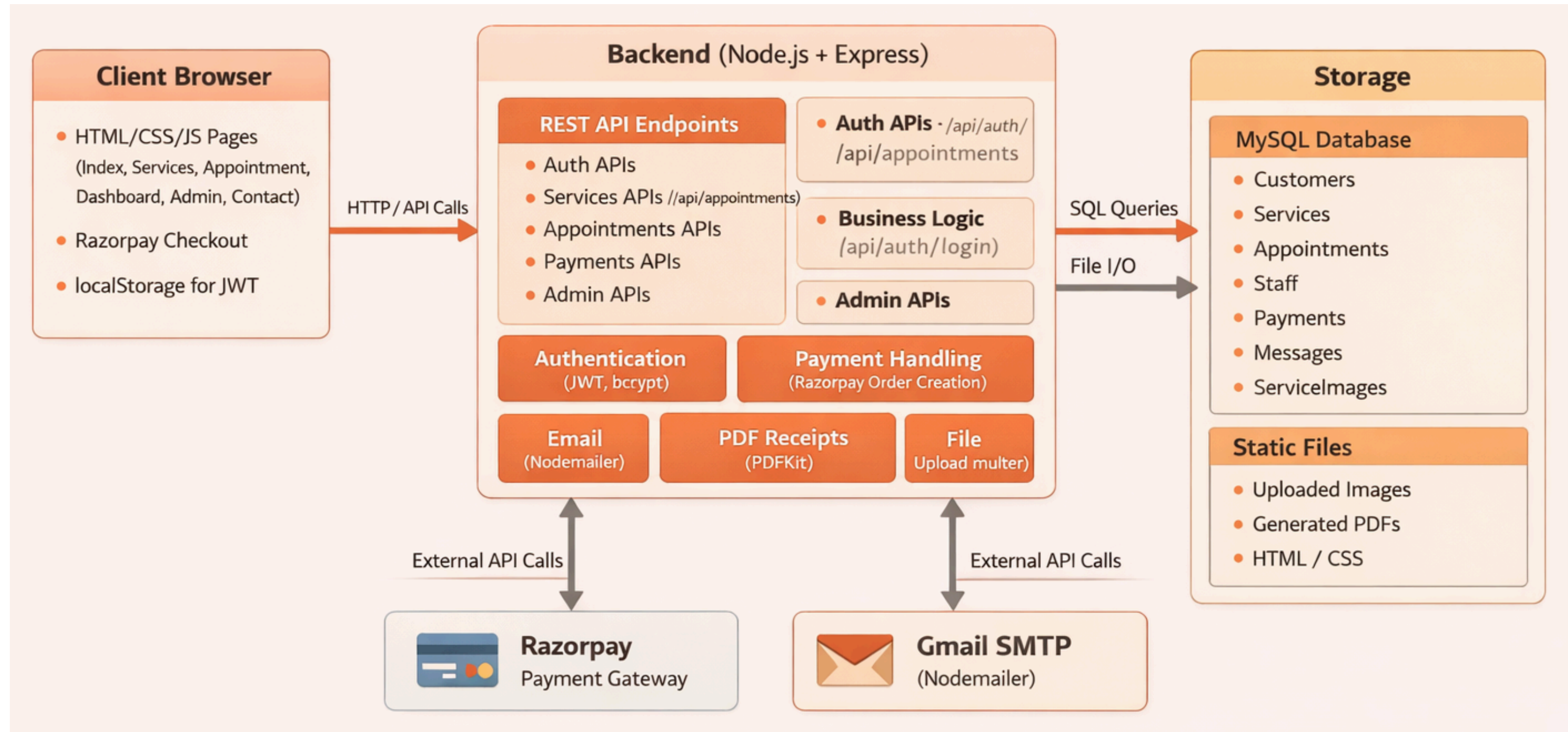
Motivation

The increasing demand for online services post-COVID-19 and the need for contactless, efficient salon operations motivated us to develop SalonSnap. We aim to digitize salon management, reduce manual workload, and provide customers with a hassle-free booking and payment experience.

Objective

- To develop a web-based salon management system that enables users to browse services and book appointments online efficiently, reducing manual scheduling efforts.
- To integrate a secure online payment gateway (Razorpay) for seamless transactions and implement automatic digital receipt generation for every successful payment.
- To design and implement an admin dashboard that allows efficient management of services, staff, appointments, customers, and payment records in a centralized system.
- To enable customer registration and authentication using secure methods, along with features for booking history tracking and receipt access.
- To implement a contact form with email notification system, allowing customers to send inquiries and receive automated responses for better communication.

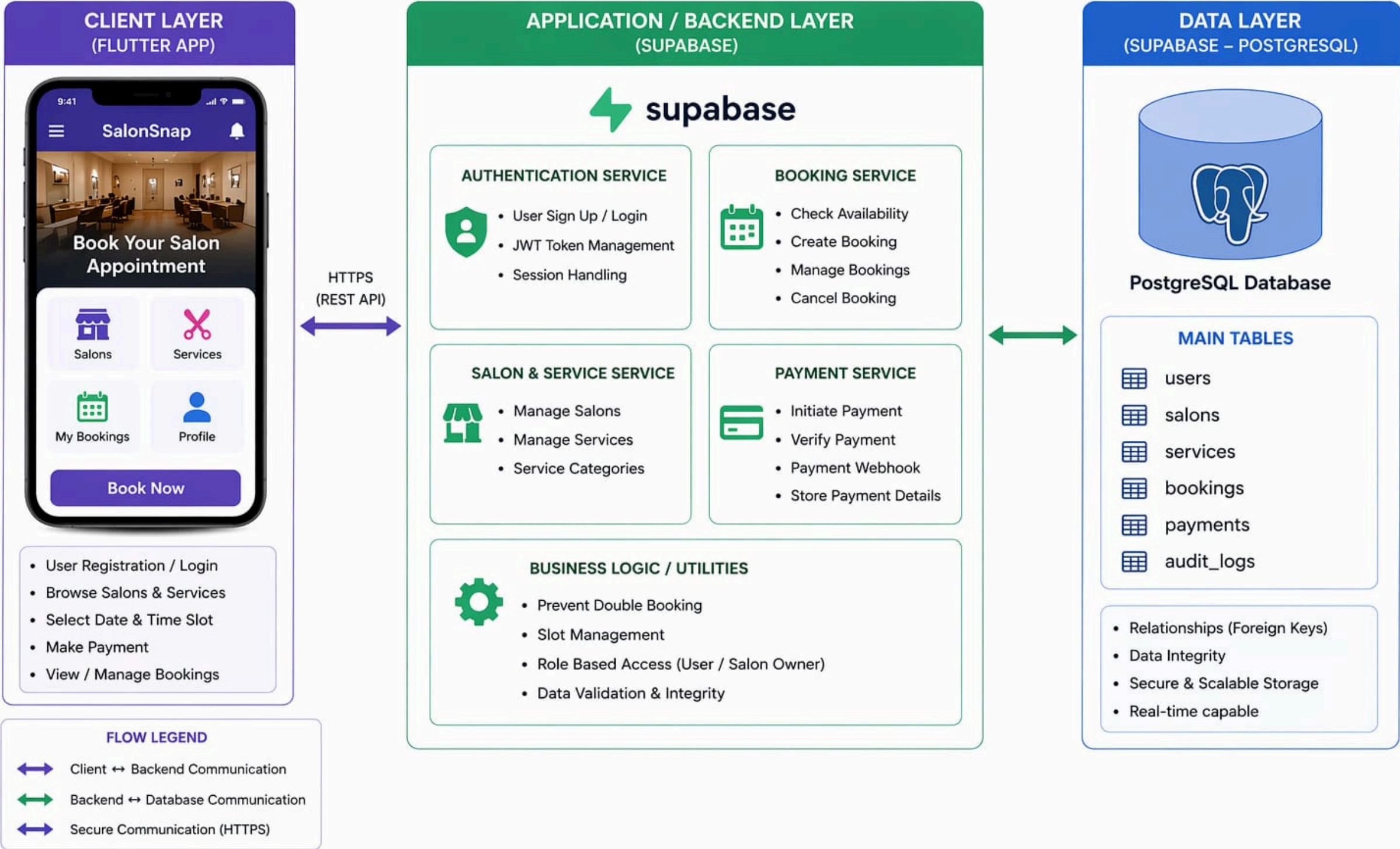
System Architecture Design



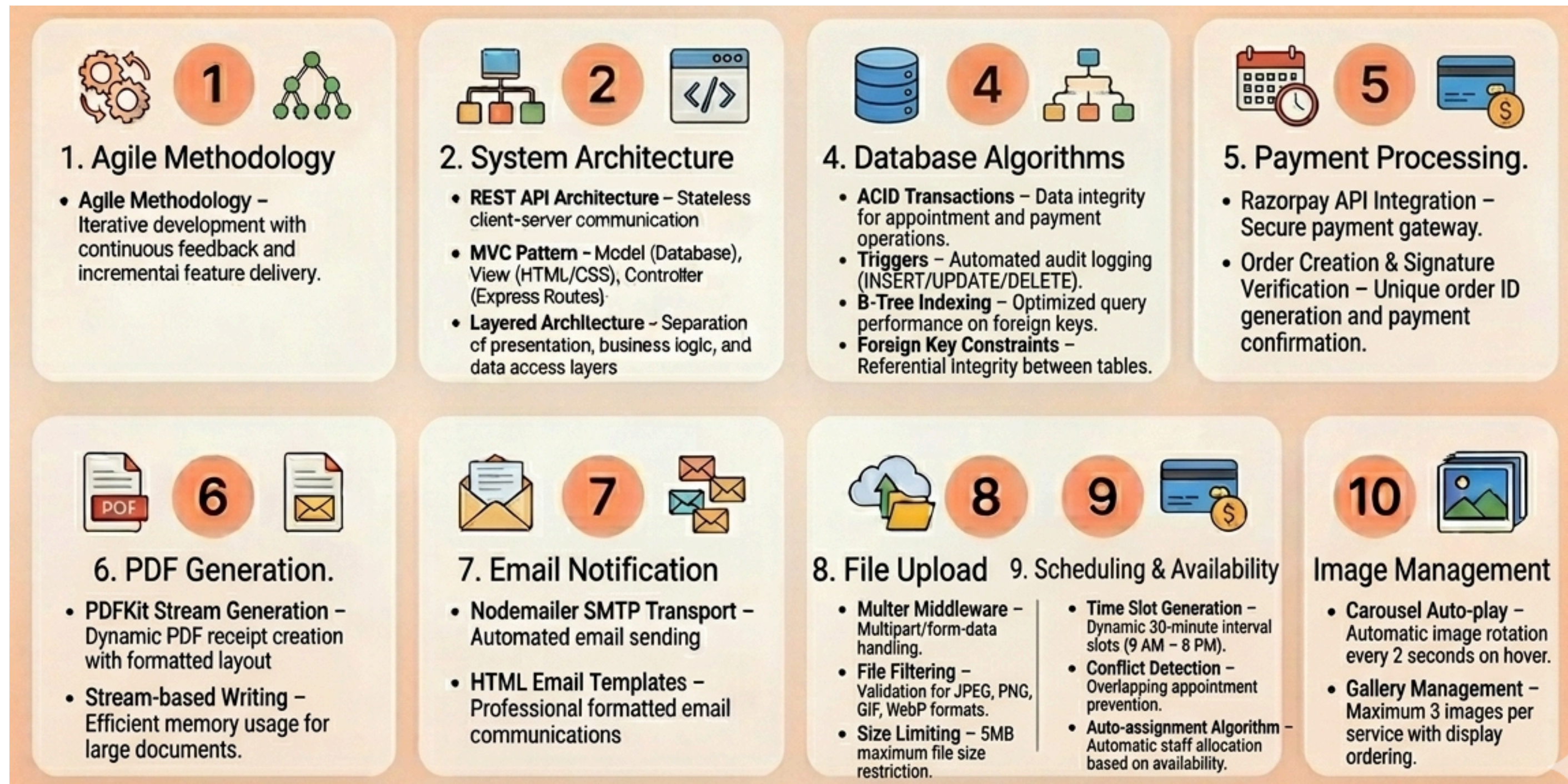
System Architecture Design

SYSTEM ARCHITECTURE – SALONSNAP

SalonSnap is a mobile based salon booking system that allows users to discover salons, book appointments, make payments and manage their bookings seamlessly.



Methodologies / Algorithms Used



Methodologies / Algorithms Used

METHODOLOGIES / ALGORITHMS USED

The following methodologies and algorithms were used in the development of **SalonSnap**.

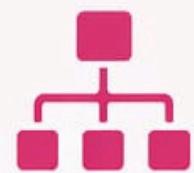
01



Agile Methodology

- Iterative development with continuous feedback.
- Features delivered in small increments.
- Regular testing and improvements.

02



System Architecture

- RESTful API architecture using Node.js + Express.
- Separation of concerns (MVC pattern).
- Communication through HTTP/HTTPS.

03



Authentication & Security

- JWT (JSON Web Token) based authentication.
- Password hashing using bcrypt.
- Role-based access control (Customer / Admin).

04



Database Design

- Relational database design (MySQL / Supabase).
- Normalized tables and relationships.
- Indexing for faster query performance.

05



Payment Processing

- Integration with Razorpay Payment Gateway.
- Secure transaction flow: Order → Payment → Verification.
- Payment status tracking.

06



PDF Generation

- Dynamic invoice and receipt generation.
- Using PDFKit library.
- Includes booking, services, tax and total amount.

07



File Upload

- Image upload for services and salon.
- Managed using Multer / Supabase Storage.
- File type and size validation.

08



Scheduling Algorithm

- Time-slot based appointment scheduling.
- Conflict detection to prevent double booking.
- Auto staff availability checking.

09



Data Validation & Optimization

- Input validation on client and server side.
- Optimized SQL queries.
- Efficient data handling for better performance.

10



Image Management

- Carousel auto play for services.
- Gallery management.
- Optimized images for faster loading.

Implementation

Module 1: Database & Backend

- MySQL
- Foreign keys + triggers (audit logging)
- Node.js + Express (port 3000, CORS, body-parser, static files)
- JWT auth (7 days) + bcrypt (10 rounds)
- dotenv config

Module 3: Service & Staff

- Services CRUD (Regular + Premium)
- Max 3 images/service (carousel auto-play)
- Staff CRUD (role, phone, email, salary)
- Staff availability check

APIs: /api/services, /api/premium, /api/staff,
/api/services-with-images

Module 2: User Management

- Customer register/login (JWT)
- Admin login + management
- Role-based access (Customer/Admin)
- Customer profile CRUD
- Admin CRUD (last admin protection)

APIs: /api/auth/*, /api/admin/*, /api/customers/:id

Module 4: Appointment System

- Book appointments (service + staff)
- 30-min slots (9 AM–8 PM)
- Conflict detection + auto/manual staff assign
- Status (Scheduled/Completed/Cancelled)
- Notes + email confirmation

APIs: /api/appointments, /api/user/appointments,
/api/available-*

Implementation

Module 5: Payment & Receipts

- Razorpay integration + cash option
- Payment status handling
- PDF receipt (PDFKit)
- Receipts stored + downloadable

APIs: /api/create-order, /api/payment-success,
/api/payments/*, /api/generate-receipt

Module 7: Contact System

- Contact form → DB storage
- Email notification to admin
- Auto-reply to customer

APIs: /api/messages, /send-message

Module 9: File & Image Handling

- Multer uploads (max 5MB)
- JPEG, PNG, GIF, WebP support
- Service image limit (max 3)
- Static file serving

Module 6: Admin Dashboard

- Dashboard stats (appointments, revenue, messages)
- Manage: Services, Staff, Customers, Appointments, Admins
- Reports + payment tracking
- Image management (upload/delete)

APIs: /api/admin/stats, /api/reports

Module 8: Security & Validation

- JWT authentication + role protection
- bcrypt password hashing
- CORS security
- Input validation + file filtering (images only)

Module 10: Scheduling & Logic

- Time slot generation (60 min)
- Conflict detection
- Auto staff assignment
- Availability tracking

Result Analysis & Validations

Full-stack salon management system successfully implemented

Core workflows: Service → Booking → Payment → Receipt

Features working correctly:

- Customer registration & login (JWT authentication)
- Service browsing (regular & premium with images)
- Appointment booking with time-slot & staff availability
- Online payment (Razorpay) + cash payment option
- PDF receipt generation & download
- Email notifications (booking + contact form)

Admin system functioning:

- Manage services, staff, customers, appointments
- View reports and messages
- Mark cash payments as paid

System behavior:

- Smooth end-to-end workflow execution
- Real-time updates in admin dashboard
- Contact form → DB storage + email working

Limitations identified:

- Admin APIs lack authentication
- JWT stored in localStorage (security risk)
- Minor performance & UI issues at scale

Database reliability:

- Foreign keys ensure data consistency
- Triggers for validation & audit logging
- Indexed queries for performance

Overall:

Reliable for small to medium-scale usage & Suitable as a working prototype/demo system

Result Analysis & Validations

System tested through manual testing of all major user flows

Customer Validation

- Registration → Account created successfully
- Login → JWT stored, dashboard access granted
- Invalid login → Error handled correctly
- Browse services → Services with images displayed
- Book appointment (cash) → Created with pending payment
- Book appointment (online) → Redirect to payment page
- Online payment → Success, receipt generated
- Dashboard → Appointments & receipts displayed correctly
- Contact form → Message saved + email sent

Database Validation

- Foreign keys → Maintain data consistency
- Triggers → Enforce validation & audit logging
- Audit tables → Track insert/update/delete actions
- Indexes → Improve query performance

Admin Validation

- Admin login → Dashboard access successful
- View appointments → All records displayed
- Mark cash payment → Status updated + receipt generated
- Manage services → Add/update/delete working
- Manage staff → CRUD operations successful
- Manage customers → View/edit/delete working
- View messages → Messages displayed & deletable
- Reports → Statistics displayed correctly

API Validation

- Auth APIs → Login/Register working (JWT)
- Service APIs → Fetch services with images
- Appointment APIs → Booking & validation working
- Payment APIs → Order creation & confirmation working
- Message APIs → Store message & send email

Result Analysis & Validations

Security Validation

- SQL Injection → Prevented (parameterised queries)
- Password Security → bcrypt hashing applied
- JWT → Secure with expiration
- CORS → Properly configured
- File Upload → Restricted (type & size)
- Admin APIs → Not protected (security gap)
- XSS Risk → Partial (innerHTML usage)

Advantage & Application

Advantages:

- 24/7 Accessibility – Customers can book appointments anytime without phone calls
- Time Efficiency – Eliminates manual booking, reduces waiting time, prevents double bookings
- Automated Receipts – Instant PDF receipt generation after successful payment
- Secure Payments – Razorpay integration ensures safe online transactions
- Centralized Management – Admin dashboard provides complete control over salon operations
- Cost Effective – Reduces administrative overhead and manual paperwork
- Customer History – Complete appointment and payment history for every customer
- Staff Management – Easy staff scheduling and performance tracking
- Email Notifications – Automated confirmations and reminders reduce no-shows
- Scalable – Can be extended to multiple salon branches
- Mobile Ready – API integration ready for mobile app connectivity
- Audit Trail – Database triggers maintain complete history of all changes

Advantage & Application

Applications:

- Beauty Salons – Haircuts, coloring, styling, makeup services
- Spa & Wellness Centers – Massage, facial, body treatment bookings
- Barbershops – Haircuts, beard trimming, shaving services
- Nail Salons – Manicure, pedicure, nail art appointments
- Fitness Centers – Class bookings, trainer sessions
- Clinics – Doctor appointment scheduling
- Hotels & Resorts – Guest spa service bookings
- Educational Institutes – Salon training center management

Conclusion

SalonSnap successfully delivers a complete, feature-rich salon management system that bridges the gap between customers and salon businesses. The system provides seamless online appointment booking with real-time availability, secure payment processing via Razorpay, automated PDF receipt generation, and a comprehensive admin dashboard for complete operational control. Key achievements include role-based authentication for customers and admins, support for regular and premium services with image carousels, dual payment methods (online and cash), JWT-based security with bcrypt password hashing, and a MySQL database with 14+ tables including triggers and audit logs. The system also features email notifications, staff management, customer history tracking, and mobile app integration via Supabase. While security gaps such as unprotected admin APIs and XSS vulnerabilities were identified, these can be mitigated with proper authentication middleware and input sanitization.

Future enhancements could include SMS notifications, multi-branch support, loyalty programs, AI-based service recommendations, WhatsApp integration, GST-compliant invoices, staff performance analytics, and inventory management.

Overall, SalonSnap transforms traditional salon management into a digital, efficient, and customer-friendly experience, making it ready for real-world deployment with a scalable foundation for future enhancements.

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Thank You